PROJECT PROPOSAL

Vessel Monitoring Dashboard

A Unified Web Platform for Real-Time Vessel Tracking, History, and Automated Alerts

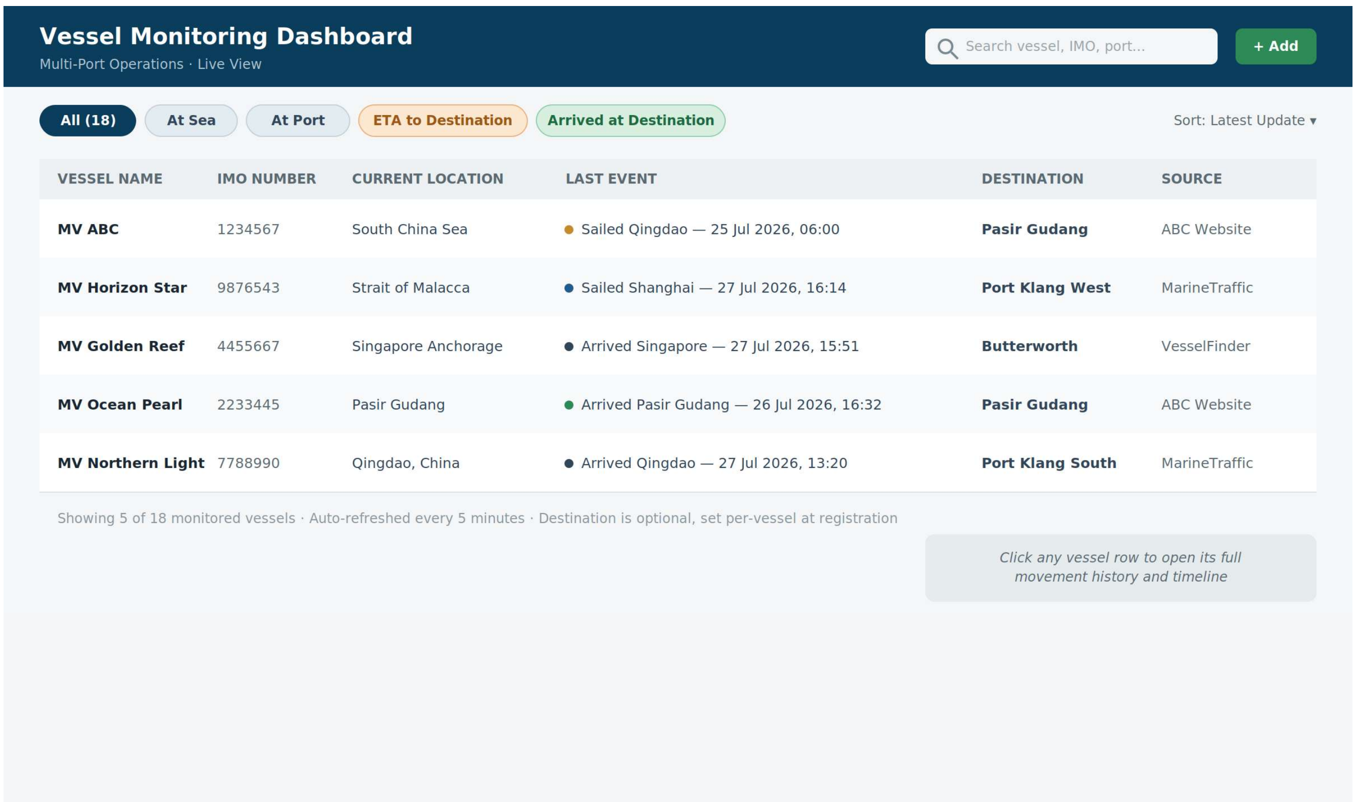

Figure 1 — Illustrative concept of the main dashboard view (final UI to be refined during design phase)

1. Executive Summary

Vessel operations currently rely on manually checking multiple third-party vessel tracking websites to determine the status, location, and estimated arrival of each ship. This is time-consuming, error-prone, and offers no centralized historical record of vessel movements.

This proposal outlines the design and delivery of a Vessel Monitoring Dashboard — a single, web-based platform that automatically aggregates vessel status from configured tracking sources, presents an always-current view of all monitored vessels, retains a complete movement history per vessel, and automatically manages the lifecycle of vessels once they arrive at their configured destination port (Pasir Gudang, Port Klang West, Butterworth, or any other port set by the user — not a single fixed destination). A companion Container/Booking Tracking module extends the same pattern to cargo-level visibility. The platform is designed to scale in a second phase to a live map view, multi-channel notifications, and AI-assisted voyage summaries, delay detection, and predictive ETAs.

The recommended stack (React/Next.js, Python FastAPI, PostgreSQL, hosted on Microsoft Azure) supports rapid delivery of the core requirements while leaving a clear path to the recommended and AI-driven features described later in this document. As this build is being delivered through AI-assisted ("vibe coding") development, the implementation timeline below is significantly shorter than a conventional manual build.

2. Project Objectives

Develop a web-based Vessel Monitoring Dashboard that allows users to monitor vessels from a single platform, removing the need to visit multiple vessel tracking websites. The platform will:

- Display the latest status of every monitored vessel in one consolidated view
- Maintain a complete, click-through historical movement record for each vessel
- Automatically retrieve and refresh vessel data from configured tracking sources
- Automatically manage vessels once they arrive at their configured destination port, and support manual removal/archiving regardless of destination
- Extend the same tracking pattern to container/booking-level cargo status, resolving what vessel-position data alone cannot show
- Provide a foundation for automated alerts and, eventually, AI-assisted insights

3. Core Requirements & Proposed Solution

3.1 Vessel Registration

Users can register a vessel for monitoring with the following fields:

- Vessel Name — e.g. MV ABC
- IMO Number — e.g. 1234567 (validated as a unique 7-digit identifier)
- **Destination Port (optional) —** selected from a configurable list of the company's common destinations (e.g. Pasir Gudang, Port Klang West, Port Klang South, Butterworth) or entered as free text for any other port. If left blank, the vessel simply continues to show its latest tracked status indefinitely — the "ETA to Destination" / "Arrived at Destination" statuses and the auto-archiving in Section 3.7 only apply once a destination is set.

The system will reject duplicate IMO numbers and confirm successful registration before the vessel appears on the dashboard.

3.2 Adding Vessels: Manual Entry & Bulk Upload

Clicking "+ Add" on the dashboard opens a two-option form: add a single vessel manually, or bulk-import many vessels at once.

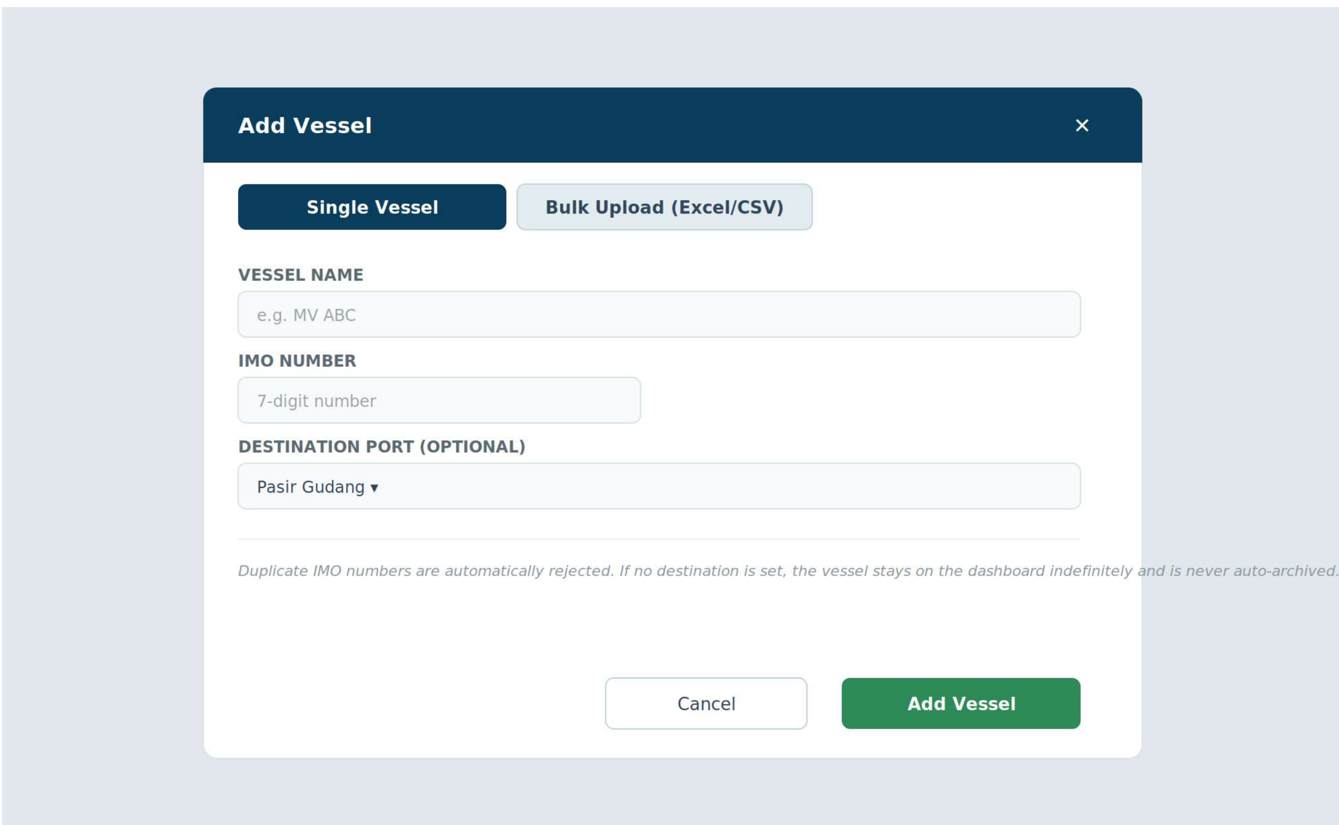

Figure 1a — Add Vessel: single-vessel manual entry (destination is optional)

For registering several vessels at once, users can upload an Excel/CSV template (Vessel Name, IMO Number, Destination Port) or a PDF vessel list. Excel/CSV is read directly, row by row. For PDF uploads, the system uses AI-assisted extraction to pull out vessel name, IMO number, and destination where present, then shows the extracted rows in an editable preview — so anything the extraction gets wrong or can't read can be corrected before anything is imported:

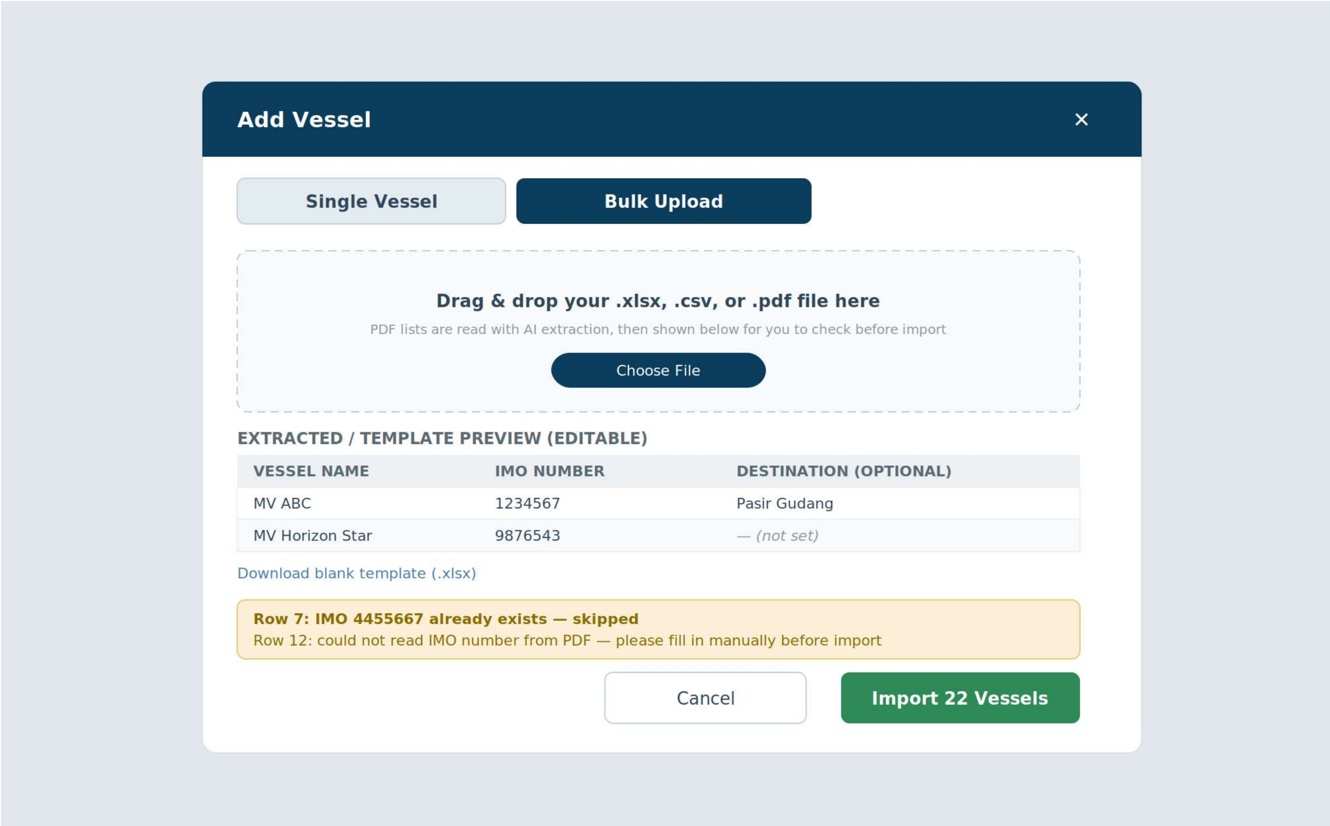

Figure 1b — Bulk Upload: Excel/CSV or PDF, with an editable preview and row-level validation before import

Excel/CSV remains the most reliable format, since the data is already structured into columns. PDF is supported for convenience, but because PDF layouts vary between sources, AI-extracted rows are always shown for review rather than imported silently — this keeps PDF upload usable without risking bad data quietly entering the dashboard.

3.3 Automated Vessel Tracking

A scheduled background service ("tracking worker") will poll each pre-configured vessel tracking website on a regular interval, extract the latest status, and update the database. Captured fields include:

- Current location
- Port of arrival / port of departure
- ETA (Estimated Time of Arrival) and ATA (Actual Time of Arrival)
- Sailing status

Each update is timestamped and tagged with its originating data source, so conflicting reports across multiple websites remain traceable.

3.3a How Status Is Detected Automatically

Each status shown on the dashboard is derived directly from what the tracking source reports — not inferred or guessed. The tracking worker compares the vessel's reported position against known ports and, if one was set, the vessel's own destination:

Figure 1c — How At Sea / At Port / ETA to Destination / Arrived at Destination / Sailed from Destination are determined

This is also why destination is optional (Section 3.1): "ETA to Destination" and "Arrived at Destination" are specifically about the vessel's own configured destination, so they simply don't apply if no destination was set — the vessel still shows Sailing / At Sea / At Port normally.

3.4 Dashboard View

All monitored vessels are shown in a single sortable table — sortable alphabetically by vessel name or by latest update — displaying vessel name, IMO number, current location, last event, destination, and data source. Rather than a separate status badge and a relative "last updated" time, the table combines both into a single "Last Event" column that states exactly what happened, where, and precisely when — e.g. "Arrived Shanghai — 20 Jul 2026, 08:00" or "Sailed Shanghai — 23 Jul 2026, 14:00" — so the vessel's most recent tracked activity is immediately clear without needing to interpret a badge. A small colour-coded dot next to the event still indicates its category (sailing, at port, arrived, etc.) for quick scanning, consistent with Section 6.E. This reflects only what tracking data can reliably confirm (see Section 3.10 below; and Section 3.3a for how each event/status is determined). The mockup below illustrates the intended layout, including filter chips and search:

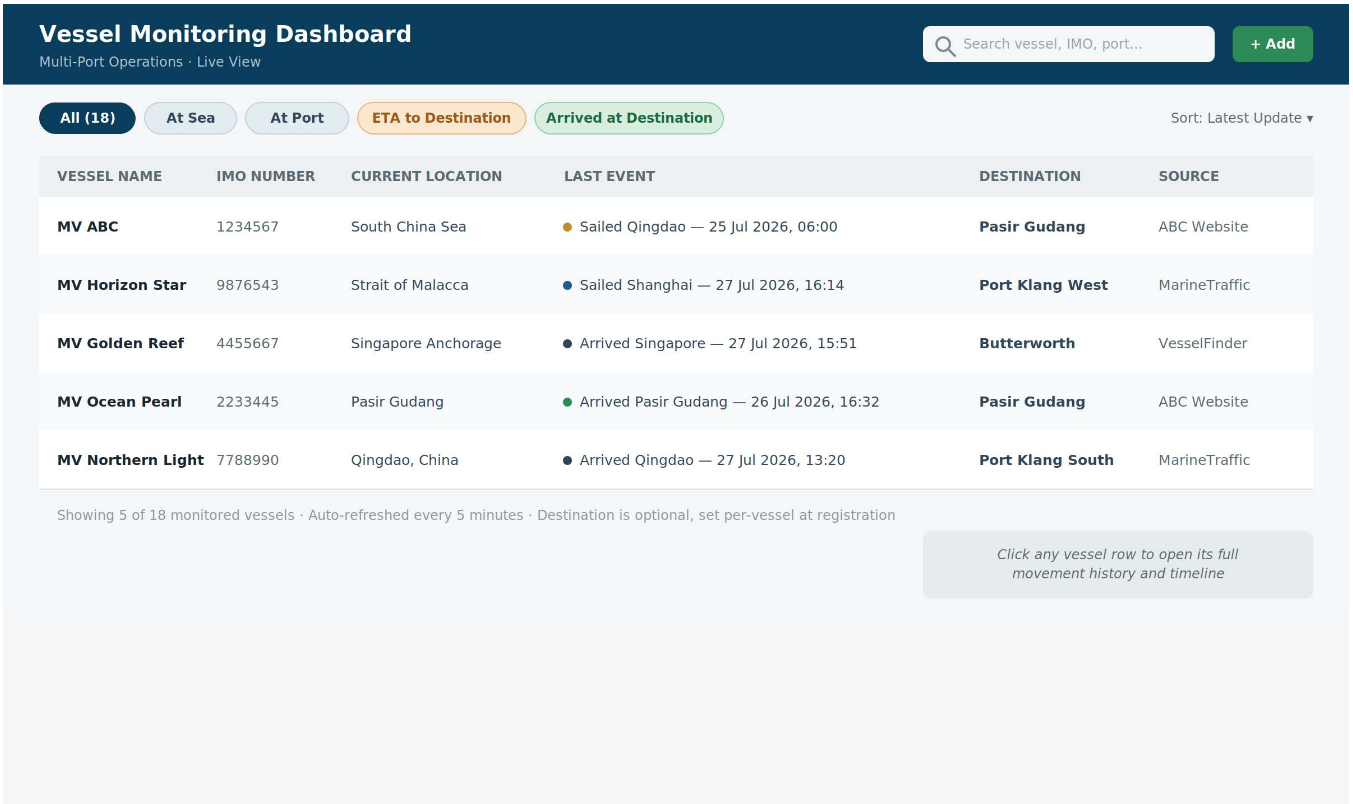

Figure 2 — Main dashboard: consolidated vessel list with Current Location and Last Event (date & time), filters, and search

3.5 Vessel History Tracking

Clicking any vessel opens its complete historical timeline, reconstructed from every status update ever recorded for that vessel — for example, arrival and departure at each port, ETA revisions, and arrival/sailing at its destination port. The current-status summary always reflects the vessel's latest event — for example, once a vessel has sailed away from its destination, the status updates to "Sailed from Destination" rather than remaining stuck on a stale ETA:

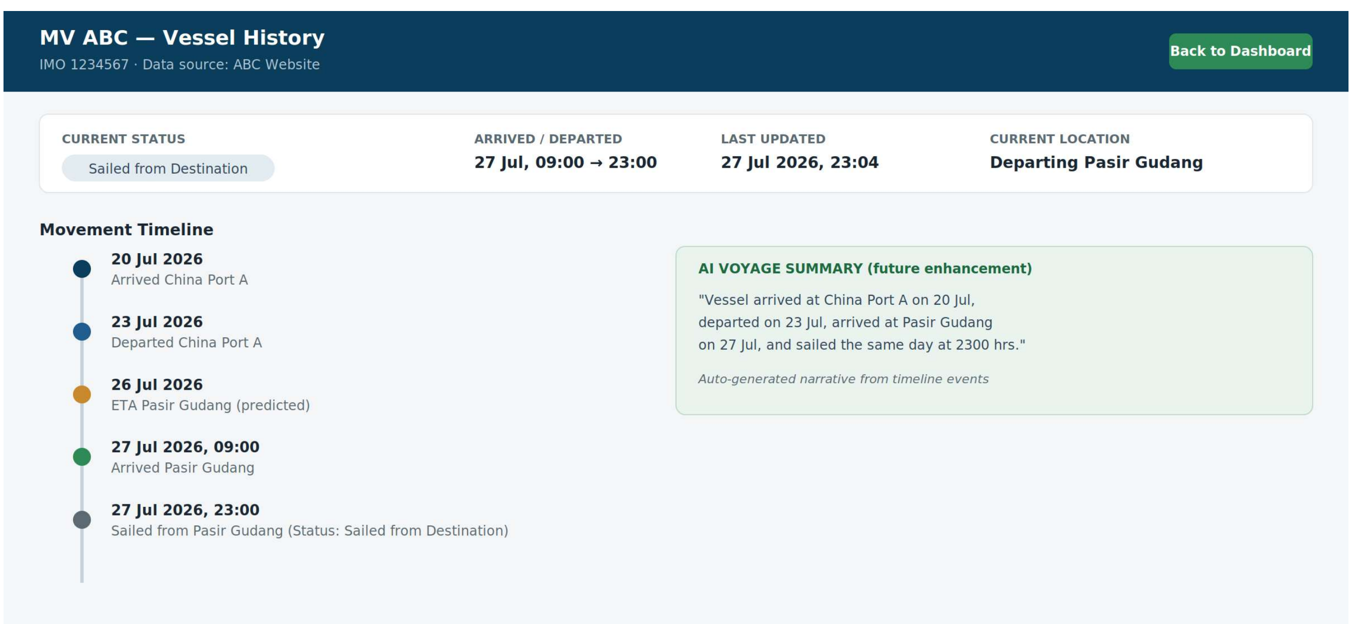

Figure 3 — Vessel history detail view with full movement timeline

3.6 Latest Status Display

The dashboard always surfaces the most recent status per vessel, while every prior update is preserved in the background for the history view — nothing is overwritten or lost.

3.7 "Arrived at Destination" Lifecycle Automation

This automation only applies to vessels that have a destination port set (Section 3.1). When a vessel's status changes to arrived at its configured destination port (whichever port was set during registration — not only Pasir Gudang), the system automatically:

- Moves the vessel into a dedicated "Arrived at Destination" section of the dashboard
- Keeps it visible in that section for a configurable retention period (default: 10 days)
- Automatically archives (or removes) it after that window, preserving its history for later reference

If no destination was set, none of this applies — the vessel simply keeps showing its latest Sailing / At Sea / At Port status indefinitely, and is never auto-archived. Regardless of destination, users can always remove a vessel manually at any point (Section 3.8) — the automation is a convenience, not a restriction.

3.8 Manual Removal

Users can stop monitoring any vessel at any time, choosing to either remove it from active monitoring or archive its history for future lookup.

3.9 Website Source Management

Admin users can add, edit, and remove vessel tracking website sources through a settings screen, so new tracking sites can be integrated in the future without a code change to the core dashboard.

3.10 On Load Port vs. Discharge Port Categorisation

Vessel tracking websites (AIS-based sources such as MarineTraffic, VesselFinder, and the Polestar GMDA asset search) report a vessel's position and port calls — they do not report what cargo activity is happening at that port. There is no reliable way to automatically label a port call as "loading" or "discharging" from this kind of data alone.

Recommendation: replace the loading/discharge auto-categorisation with a neutral, verifiable status — "At Sea", "At Port: [port name]", "ETA to Destination", "Arrived at Destination" — as reflected in the dashboard mockup (Figure 2). If load/discharge activity needs to be tracked, it is more reliably obtained from the shipping line's own booking/cargo record (e.g. ONE's eCommerce booking portal), which explicitly tracks Booking Confirmed → Loaded → In Transit → Discharged → Gate Out. This is exactly what the Container/Booking Tracking module in Section 4 is designed to provide.

4. Container / Booking Tracking Module

To directly address cargo-level visibility that vessel position data cannot provide, we recommend a companion module — structured the same way as the vessel dashboard — that tracks container numbers or booking numbers against the shipping line's own booking/cargo systems (e.g. ONE, Maersk, MSC, CMA CGM, InterAsia — see Section 8.1 for the specific tracking pages).

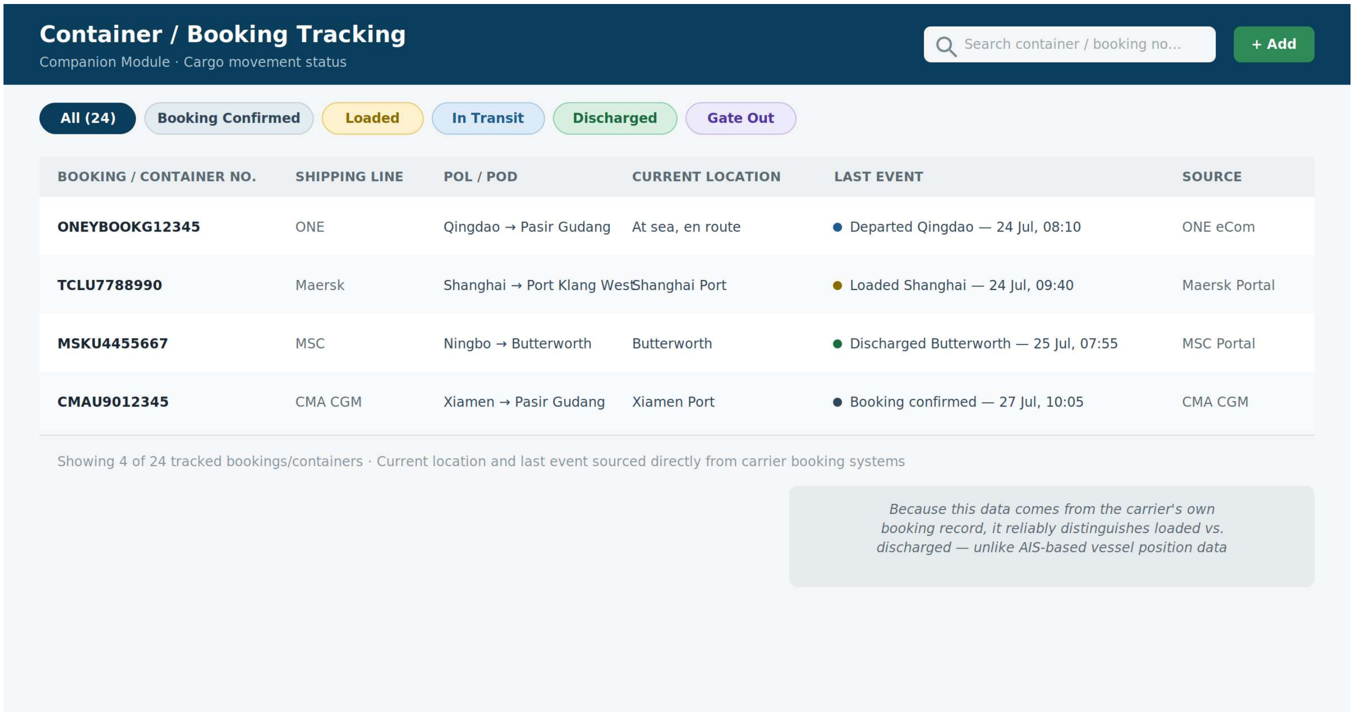

Figure 3a — Container/Booking tracking table: Current Location and Last Event sourced from carrier booking records

Consistent with the vessel dashboard (Section 3.4), this table also uses a Current Location + Last Event format rather than a status badge — e.g. "Loaded Shanghai — 24 Jul 2026, 09:40" or "Discharged Butterworth — 25 Jul 2026, 07:55" — so it's immediately clear where the cargo is and exactly what last happened to it. Because this data originates from the carrier's own booking record rather than AIS position reports, it reliably distinguishes Booking Confirmed, Loaded, In Transit, Discharged, and Gate Out events — resolving the load/discharge visibility gap described in Section 3.10. Columns include:

- Booking/Container Number
- Shipping Line
- Port of Loading (POL) and Port of Discharge (POD)
- Current Location
- Last Event (event type, port, and exact date/time — e.g. Loaded, Discharged, Booking Confirmed)
- Data source

This module can share the same search, filter, and colour-coding patterns as the vessel dashboard, and can be delivered as an additional phase once the vessel tracking core is stable.

5. End Goal

A single dashboard providing a complete, always-current view of all monitored vessels — current status, historical movements, ETA updates, arrival records, and automated alerts — eliminating the need to manually check multiple vessel tracking websites.

6. Recommended Features

6.A Search Function

Users can search the vessel list by vessel name, IMO number, port, or destination, narrowing the dashboard table instantly as they type.

6.B Map View

A live map shows every active vessel's current position, route, and destination — supporting all configured destination ports (Pasir Gudang, Port Klang West, Butterworth, and others), not a single fixed port:

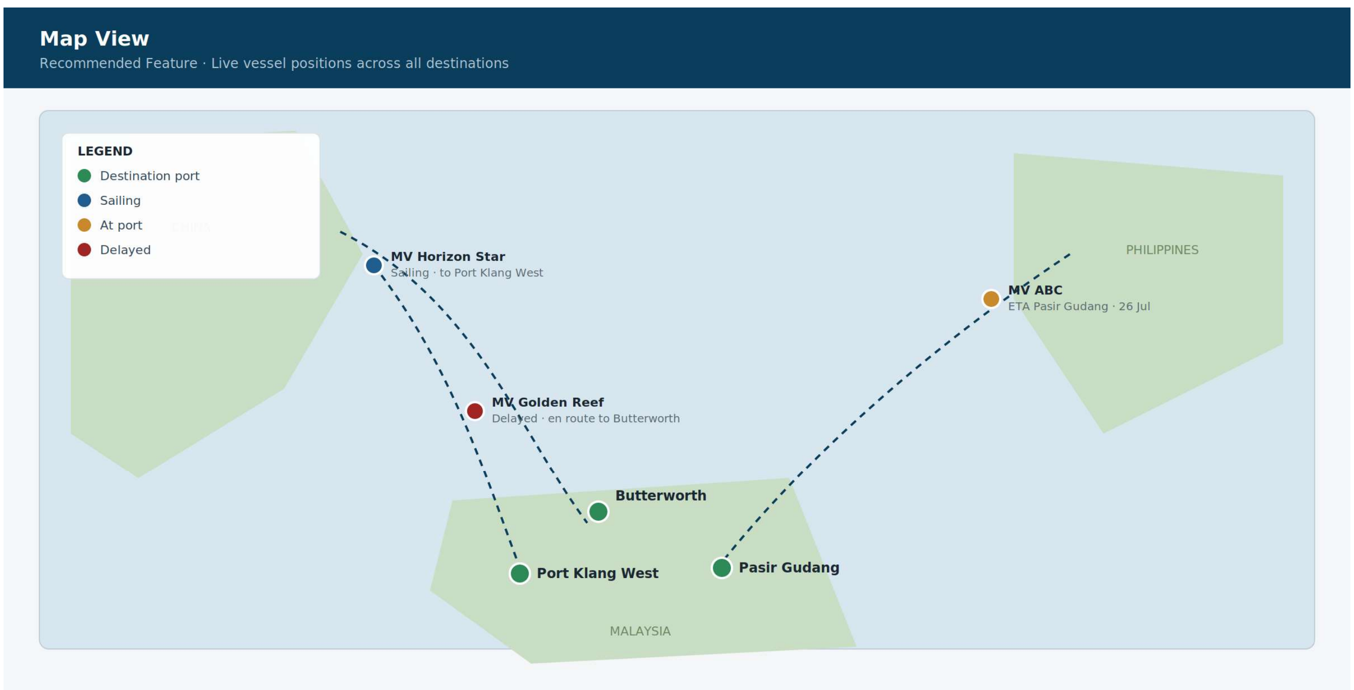

Figure 4 — Map view concept: vessel positions, routes, and destination markers

6.C Notifications

Automated notifications are triggered on vessel arrival, vessel departure, ETA changes, delays, and arrival at destination. Planned channels are email and Microsoft Teams at launch, with WhatsApp planned as a future enhancement:

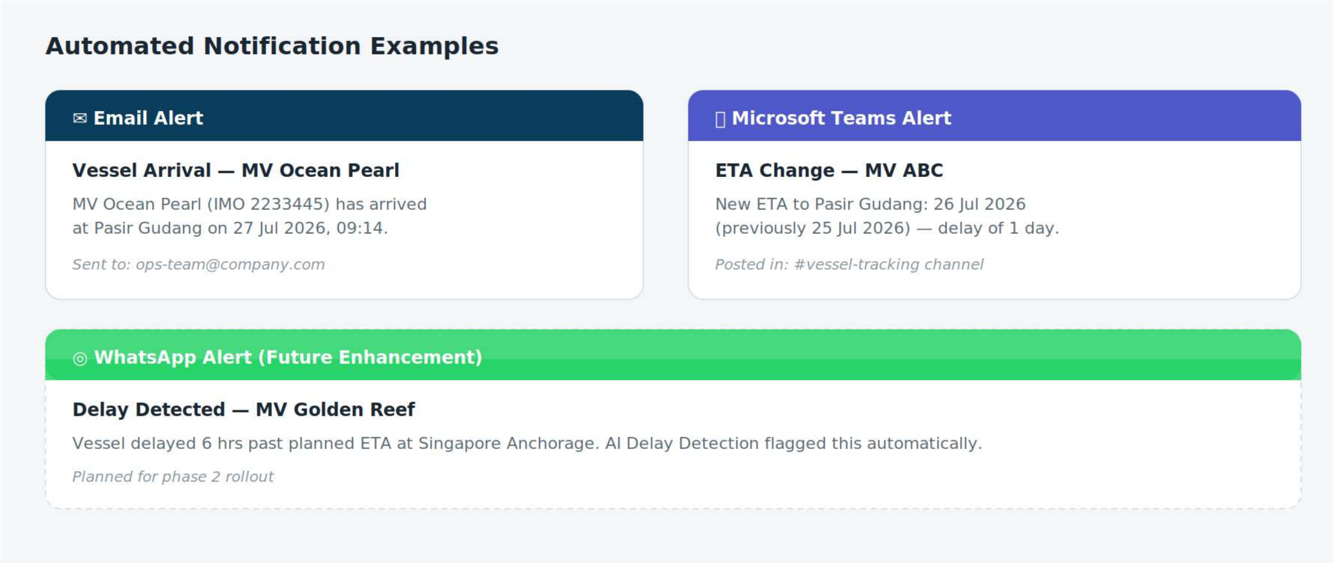

Figure 5 — Notification examples across email, Microsoft Teams, and (future) WhatsApp

6.D Dashboard Filters

Quick filters let users narrow the vessel list to: At Sea, At Port, ETA to Destination, or Arrived at Destination (shown as filter chips in Figure 2) — matching the neutral, verifiable statuses described in Section 3.10, rather than a fixed "Pasir Gudang" filter.

6.E Status Colour Coding

Since the dashboard shows Last Event text rather than a status badge (Section 3.4), the colour coding appears as a small dot next to each Last Event entry, letting users still scan for a category at a glance:

Colour	Meaning
Green	Arrived
Blue	Sailing
Orange	At Port
Red	Delayed

7. AI Features (Future Enhancement)

- **AI Voyage Summary —** automatically generates a plain-language narrative of a vessel's voyage, e.g. "Vessel arrived at China Port A on 20 Jul, departed on 23 Jul, arrived at Pasir Gudang on 27 Jul, and sailed the same day at 2300 hrs."
- **AI Delay Detection —** compares planned ETA against actual arrival and automatically highlights delays.
- **AI Predictive ETA —** uses vessel speed, route, and historical data to refine arrival predictions beyond the source website's ETA.
- **AI Exception Alerts —** automatically flags delays, route deviations, unexpected port calls, and unusually long port stays.
- **Reports —** generates daily reports covering active vessels, ETA to destination, delayed vessels, and arrived vessels, exportable to Excel or PDF.

8. Recommended Technology

Layer	Technology	Rationale
Frontend	React / Next.js	Fast, component-driven UI; server-side rendering for quick initial load of the dashboard table.
Backend	Python FastAPI	High-performance async API, ideal for scheduled scraping/polling jobs and REST endpoints.
Database	PostgreSQL	Reliable relational storage for vessels, status history, and audit trail of updates.
AI	Azure OpenAI / OpenAI API	Powers voyage summaries, delay detection, and predictive ETA in the future enhancement phase.
Hosting	Microsoft Azure	Managed hosting, scheduled jobs (Azure Functions), and straightforward integration with Microsoft Teams notifications.

8.1 Example Tracking Data Sources

For reference, the tracking worker (Section 3.3) is designed to poll sources such as:

- **Vessel position/status —** MarineTraffic, VesselFinder, and the Polestar GMDA asset search, plus any vessel operator's own tracking website.

Container/booking status — carrier tracking portals, including:

- **InterAsia —** https://www.interasia.cc/Service/Form?servicetype=1
- **CMA CGM —** https://www.cma-cgm.com/eBusiness/Tracking
- **Maersk —** https://www.maersk.com/tracking/
- **MSC —** https://www.msc.com/en/track-a-shipment
- **ONE (Ocean Network Express) —** https://ecomm.one-line.com/one-ecom/manage-shipment/cargo-tracking

These feed the Current Location and Last Event fields in the Container/Booking Tracking module (Section 4), the same way vessel-tracking sites feed the vessel dashboard.

9. Proposed Implementation Phases

This project will be delivered using an AI-assisted ("vibe coding") development approach, where features are built and iterated directly with AI coding tools rather than through traditional manual sprint estimation. This significantly compresses the timelines typical of conventional development while keeping the same phase structure and scope below.

Phase	Scope	Indicative Duration
Phase 1	Vessel registration (destination optional), manual + bulk add (Excel/CSV/PDF with AI extraction review), automated tracking (initial sources), dashboard view, history timeline, latest-status display	4–6 days
Phase 2	Arrived-at-PG automation, manual removal/archiving, website source management (admin)	2–3 days
Phase 3	Search, dashboard filters, colour coding, map view	2–3 days
Phase 4	Notifications (email, Microsoft Teams), reports (Excel/PDF export)	1–2 days
Phase 5	Container/Booking Tracking module (Section 4): booking/container table, carrier source integration (e.g. ONE eCommerce), shared search/filter patterns	2–3 days
Phase 6 (Future)	AI voyage summary, delay detection, predictive ETA, exception alerts, WhatsApp notifications	To be scoped

Total estimated build time for Phases 1–5: approximately 2.5–3.5 weeks (Phase 1 slightly extended to cover bulk/PDF import), subject to timely access to tracking-website/booking-portal sources and stakeholder feedback turnaround during review checkpoints.

10. Assumptions & Next Steps

- Access credentials/API terms for each vessel tracking website (e.g. MarineTraffic, VesselFinder) will be confirmed and made available before Phase 1 begins.
- Polling frequency will be tuned to balance data freshness against each source website's rate limits/terms of use.
- Final visual design (branding, colours, layout) will be confirmed with stakeholders during a short UI design sprint; figures in this proposal are illustrative concepts, not final UI.
- Microsoft Teams and email notification channels require organisational admin access to configure; WhatsApp integration is scoped for a later phase.
- Because the build is AI-assisted, fast stakeholder feedback at each phase checkpoint is important to keep the compressed timeline on track — delayed reviews will extend delivery.
- The initial list of destination ports (Pasir Gudang, Port Klang West, Port Klang South, Butterworth) will be confirmed with stakeholders and configured as selectable options, with free-text entry for any additional port.
- Access to carrier booking portals (e.g. ONE eCommerce) for the Container/Booking Tracking module will need to be confirmed — either via existing account access or a scraping-compatible interface — before Phase 5 begins.
- AI-assisted extraction from PDF vessel lists (Section 3.2) works best on reasonably clean, text-based PDFs; scanned or heavily formatted PDFs may need more manual correction in the review step.
- Destination Port is optional by design (Section 3.1) — vessels without one are tracked and displayed like any other, just without the arrival/auto-archive automation.

We recommend a short kickoff workshop to confirm tracking website sources, notification recipients, and Phase 1 priorities before development begins.